Question ID: afec1a70

Question

While researching a topic, a student has taken the following notes:

- As engineered structures, many bird nests are uniquely flexible yet cohesive.
- A research team led by Yashraj Bhosale wanted to better understand the mechanics behind these structural properties.
- Bhosale's team used laboratory models that simulated the arrangement of flexible sticks into nest-like structures.
- The researchers analyzed the points where sticks touched one another.
- When pressure was applied to the model nests, the number of contact points between the sticks increased, making the structures stiffer.

The student wants to present the primary aim of the research study. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

A. Bhosale's team wanted to better understand the mechanics behind bird nests' uniquely flexible yet cohesive structural properties.

B. The researchers used laboratory models that simulated the arrangement of flexible sticks and analyzed the points where sticks touched one another.

C. After analyzing the points where sticks touched, the researchers found that the structures became stiffer when pressure was applied.

D. As analyzed by Bhosale's team, bird nests are uniquely flexible yet cohesive engineered structures.

Question ID: 6c9df5d1

Question

While researching a topic, a student has taken the following notes:

- Some powerful works of literature have so influenced readers that new legislation has been passed as a result.
- *The Interesting Narrative of the Life of Olaudah Equiano* (1789) is the autobiography of a man who endured slavery on both sides of the Atlantic.
- Equiano's book contributed to the passage of the Slave Trade Act of 1807.
- *The Jungle* (1906) is a fictional work by Upton Sinclair that describes unsanitary conditions in US meatpacking plants.
- Sinclair's book contributed to the passage of the Pure Food and Drug Act in 1906.

The student wants to emphasize a difference between the two books. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

A. Although both are powerful works of literature that contributed to new legislation, Equiano's book is an autobiography, while Sinclair's is fictional.

B. They may have written about different topics, but Equiano and Sinclair both influenced readers.

C. The 1807 Slave Trade Act resulted in part from a book by Equiano, while the 1906 Pure Food and Drug Act resulted in part from a book by Sinclair.

D. *The Interesting Narrative of the Life of Olaudah Equiano* and *The Jungle* are two works of literature that contributed to new legislation (concerning the slave trade and food safety, respectively).

Question ID: 16631d34

Question

While researching a topic, a student has taken the following notes:

- The Million Song Dataset (MSD) includes main audio features and descriptive tags for popular songs.
- Audio features include acoustic traits such as loudness and pitch intervals.
- Many algorithms use these audio features to predict a new song's popularity.
- These algorithms may fail to accurately identify main audio features of a song with varying acoustic traits.
- Algorithms based on descriptive tags that describe fixed traits such as genre are more reliable predictors of song popularity.

The student wants to explain a disadvantage of relying on audio features to predict a song's popularity. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

A. Many popularity-predicting algorithms are based on a song's audio features, such as loudness and pitch intervals.

B. Algorithms based on audio features may misidentify the main features of a song with varying acoustic traits, making such algorithms less reliable predictors of popularity than those based on fixed traits.

C. Audio features describe acoustic traits such as pitch intervals, which may vary within a song, whereas descriptive tags describe fixed traits such as genre, which are reliable predictors of popularity.

D. The MSD's descriptive tags are reliable predictors of a song's popularity, as the traits they describe are fixed.

Question ID: b07a7634

Question

While researching a topic, a student has taken the following notes:

- Digital Light Synthesis (DLS) is a form of additive manufacturing that utilizes light to rapidly cure liquid resin into high-quality, 3D objects.
- Step 1: Ultraviolet (UV) light images are projected up into a pool of liquid resin, where the object's first layer takes shape.
- Step 2: The partially cured resin object is raised, leaving a thin space (a "dead zone") beneath it for oxygen and liquid resin to flow through.
- Step 3: The UV light passes through the dead zone—maintaining the flow of resin—and partially cures additional layers of the object.
- Step 4: When the resin object is complete, it is baked in an oven to complete the curing.

The student wants to describe how DLS cures 3D objects. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

A. DLS is a form of additive manufacturing that creates a "dead zone" in which UV light solidifies layer by layer before being baked in an oven, creating a high-quality, 3D object.

B. DLS cures 3D objects by passing through a "dead zone," adding layers to the object, then curing the object in an oven.

C. In DLS, UV light images are projected into a liquid resin pool to cure a 3D object layer by layer; once solidified, the object is baked in an oven.

D. In DLS, UV light is projected into layers of liquid resin until the resin solidifies and passes through a "dead zone," wherein the curing is completed.

Question ID: 3fa48bf3

Question

While researching a topic, a student has taken the following notes:

- British scholar Robert Plot described fossilized dinosaur bones in his 1676 book *The Natural History of Oxfordshire*.
- Plot earned a reputation for being the first person to have discovered dinosaur remains.
- In 1990, archaeologists in Lesotho, in southern Africa, discovered a fossilized phalanx of a *Massospondylus carinatus* dinosaur in a cave once inhabited by humans.
- Indigenous Khoesan and Basotho peoples had inhabited the cave beginning around 1100 CE.
- According to paleontologist Julien Benoit, these peoples may have found the phalanx and brought it to the cave centuries before Plot's descriptions.

The student wants to emphasize the significance of the 1990 discovery to Plot's reputation. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

A. Benoit challenged Plot's reputation for being the first person to have discovered *M. carinatus* remains.

B. Evidence that Khoesan and Basotho peoples may have found an *M. carinatus* phalanx as long ago as 1100 CE suggests that Plot may not have been the first person to have discovered dinosaur remains.

C. According to Benoit's analysis of the 1990 discovery, Indigenous peoples in southern Africa may have brought the fossilized phalanx to the cave as long ago as 1100 CE.

D. In 1990, more than three centuries after Plot claimed in his book that he had found fossilized dinosaur bones, archaeologists uncovered evidence in southern Africa that disproved his claims.

Question ID: 8fe4f4ab

Question

While researching a topic, a student has taken the following notes:

- One of history's greatest libraries was the House of Wisdom in Baghdad, Iraq.
- It was founded in the eighth century with the goal of preserving all the world's knowledge.
- Scholars at the House of Wisdom collected ancient and contemporary texts from Greece, India, and elsewhere and translated them into Arabic.
- Writings included those of the Greek philosopher Aristotle and the Indian mathematician Aryabhata.
- The House of Wisdom used Chinese papermaking technology to create paper versions to be studied and shared.

The student wants to explain how the House of Wisdom preserved the world's knowledge. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

A. The House of Wisdom was known for bringing together knowledge from around the world, including from Greece, India, and China.

B. Founded in Iraq in the eighth century, the House of Wisdom employed many scholars as translators.

C. Writings from the Greek philosopher Aristotle and the Indian mathematician Aryabhata were preserved at the House of Wisdom.

D. The House of Wisdom collected writings from different countries and created paper versions in Arabic to be studied and shared.

Question ID: 964c6055

Question

While researching a topic, a student has taken the following notes:

- Two opposing theories of vision divided scholars for many centuries.
- The ancient Greek mathematician Euclid (circa 300 BCE) supported the extramission theory.
- This theory held that the eyes emit a form of radiation that illuminates objects in its range.
- The ancient Greek philosopher Aristotle (384–322 BCE) supported the intromission theory.
- This theory held that objects emit a form of radiation that reaches the eyes.
- In the eleventh century, Arab mathematician Ibn al-Haytham (965–1040 CE) largely settled the debate with the first conclusive experiments supporting intromission.

The student wants to provide a historical overview of the two theories. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

A. Scholars were divided between the extramission and intromission theories of vision until Ibn al-Haytham's eleventh-century experiments largely settled the debate in support of intromission.

B. Through two opposing theories of vision—extramission and intromission—Euclid, Aristotle, and Ibn al-Haytham held that a form of radiation is emitted either from objects or from the eyes.

C. While Ibn al-Haytham largely settled the debate in the eleventh century, Aristotle supported the theory of intromission centuries before.

D. Before the eleventh century, the ancient Greek philosopher Aristotle supported the intromission theory, which held that objects emit a form of radiation that reaches the eyes.

Question ID: 64e88c58

Question

While researching a topic, a student has taken the following notes:

- In 1971, experimental musician Pauline Oliveros created *Sonic Meditations*.
- *Sonic Meditations* is not music but rather a series of sound-based exercises called meditations.
- Each meditation consists of instructions for participants to make, imagine, listen to, or remember sounds.
- The instructions for Meditation V state, “walk so silently that the bottoms of your feet become ears.”
- Those for Meditation XVIII state, “listen to a sound until you no longer recognize it.”

The student wants to provide an explanation and an example of Oliveros’s *Sonic Meditations*. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

A. *Sonic Meditations* is not music but rather a series of sound-based meditations that consist of instructions; Meditation XVIII, for instance, instructs participants to “listen to a sound until you no longer recognize it.”

B. In 1971, Oliveros created *Sonic Meditations*, a series of meditations that consist of instructions for participants to make, imagine, listen to, or remember sounds.

C. “Walk so silently that the bottoms of your feet become ears” is one example of the instructions found in Oliveros’s *Sonic Meditations*.

D. While both meditations consist of instructions, Meditation XVIII instructs participants to “listen,” whereas Meditation V instructs participants to “walk.”

Question ID: af88c47a

Question

While researching a topic, a student has taken the following notes:

- Freddie Wong (born 1985) is a director and special effects artist from the United States.
- He is best known for the action-comedy web series *Video Game High School* (*VGHS*).
- *VGHS* premiered in 2012 on RocketJump, a YouTube channel that Wong cocreated.
- The series was celebrated for its inventive video game–centric world and high-quality special effects.
- *VGHS* was nominated for a Producers Guild Award for Outstanding Digital Series.

The student wants to begin a narrative about Wong's award-nominated web series. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

A. In 2012, director and visual effects artist Freddie Wong launched a new action-comedy web series: *Video Game High School*.

B. *Video Game High School* was celebrated for its inventive video game–centric world and high-quality special effects, and it was nominated for a Producer's Guild Award for Outstanding Digital Series.

C. Wong, cocreator of the YouTube channel RocketJump, would go on to see his web series be nominated for a Producers Guild Award.

D. In 2012, *Video Game High School* premiered on RocketJump; it would later be nominated for an award.

Question ID: 5fa51c86

Question

While researching a topic, a student has taken the following notes:

- Ulaanbaatar is the capital of Mongolia.
- The city's population is 907,802.
- Ulaanbaatar contains 31.98 percent of Mongolia's population.
- Hanoi is the capital of Vietnam.
- The city's population is 7,781,631.
- Hanoi contains 8.14 percent of Vietnam's population.

The student wants to emphasize the relative sizes of the two capitals' populations. Which choice most effectively uses information from the given sentences to emphasize the relative sizes of the two capitals' populations?

Answer

A. Mongolia's capital is Ulaanbaatar, which has 907,802 people, and Vietnam's capital is Hanoi, which has 7,781,631 people.

B. Comparing Vietnam and Mongolia, 7,781,631 is 8.14 percent of Vietnam's population, and 907,802 is 31.98 percent of Mongolia's.

C. Even though Hanoi (population 7,781,631) is larger than Ulaanbaatar (population 907,802), Ulaanbaatar accounts for more of its country's population.

D. The populations of the capitals of Mongolia and Vietnam are 907,802 (Ulaanbaatar) and 7,781,631 (Hanoi), respectively.

Question ID: fdd9a360

Question

While researching a topic, a student has taken the following notes:

- The popular wood-wide web theory posits that trees can communicate and exchange resources with one another via common mycorrhizal networks (CMNs) of fungi.
- Ecologist Dr. Suzanne Simard first suggested this theory in 1997.
- She described trees as “super-cooperators.”
- In the 2022 study “The Decay of the Wood-Wide Web?,” mycologist Dr. Justine Karst and colleagues evaluated dozens of CMN studies.
- They write that CMNs “have captured the interest of broad audiences. We are concerned, however, that recent claims about CMNs in forests are disconnected from evidence.”

The student wants to use a quotation to emphasize a potential problem with the wood-wide web theory. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

A. Describing trees as “super-cooperators,” Simard first suggested that trees can exchange resources with one another in 1997.

B. In “The Decay of the Wood-Wide Web?,” Karst and colleagues note that common mycorrhizal networks “have captured the interest of broad audiences.”

C. After evaluating dozens of CMN studies, Karst and colleagues expressed concern that recent claims about common mycorrhizal networks are “disconnected from evidence.”

D. Despite the concerns expressed in the 2022 study “The Decay of the Wood-Wide Web?,” the wood-wide web theory remains popular.

Question ID: 10cd0327

Question

While researching a topic, a student has taken the following notes:

- A thermal inversion is a phenomenon where a layer of atmosphere is warmer than the layer beneath it.
- In 2022, a team of researchers studied the presence of thermal inversions in twenty-five gas giants.
- Gas giants are planets largely composed of helium and hydrogen.
- The team found that gas giants featuring a thermal inversion were also likely to contain heat-absorbing metals.
- One explanation for this relationship is that these metals may reside in a planet's upper atmosphere, where their absorbed heat causes an increase in temperature.

The student wants to present the study's findings to an audience already familiar with thermal inversions. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

A. Heat-absorbing metals may reside in a planet's upper atmosphere.

B. The team studied thermal inversions in twenty-five gas giants, which are largely composed of helium and hydrogen.

C. Researchers found that gas giants featuring a thermal inversion were likely to contain heat-absorbing metals, which may reside in the planets' upper atmospheres.

D. Gas giants were likely to contain heat-absorbing metals when they featured a layer of atmosphere warmer than the layer beneath it, researchers found; this phenomenon is known as a thermal inversion.

Question ID: 5b8b69a2

Question

While researching a topic, a student has taken the following notes:

- Archaeologist Jon Erlandson and colleagues argue that humans first arrived in the Americas by sea.
- They propose that humans traveled between Pacific Ocean islands and coastlines from northeast Asia to the Americas.
- Many of these islands and coastal zones were later submerged as glaciers melted and sea levels rose.
- The researchers think that “a coastal route, including kelp forests and estuaries, would have provided a rich mix of marine, estuarine, riverine, and terrestrial resources” such as seaweeds, fish, and birds.
- This proposed scenario is known as the kelp highway hypothesis.

The student wants to summarize the kelp highway hypothesis. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

A. Pacific Ocean islands and coastlines likely contained “a rich mix of marine, estuarine, riverine, and terrestrial resources” such as seaweeds, fish, and birds, according to researchers.

B. One argument about how humans first arrived in the Americas is the kelp highway hypothesis proposed by Jon Erlandson and colleagues.

C. Humans may have first arrived in the Americas by sea, traveling between Pacific Ocean islands and coastlines and subsisting on a variety of resources.

D. As glaciers melted and sea levels rose, many Pacific Ocean islands and coastal zones were submerged.

Question ID: 5222ffab

Question

While researching a topic, a student has taken the following notes:

- Neuroscientists Krishnan Padmanabhan and Zhen Chen sought to better understand the workings of the brain's olfactory system.
- They devised a study using mathematical models.
- They found that certain fibers allow the brain to toggle from one method of processing smells to another.
- In one method, cells in the piriform cortex (where the perception of odor forms) capture olfactory information at a given moment.
- In the other, the cells track changes in olfactory information over time.

The student wants to summarize the study's findings. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

A. To arrive at these findings, which describe dual methods of processing smells in the piriform cortex, Padmanabhan and Chen devised a study using mathematical models.

B. Padmanabhan and Chen showed that olfactory information is captured by cells in the piriform cortex, where the perception of odor forms.

C. Using mathematical models, Padmanabhan and Chen devised a study to better understand the workings of the brain's olfactory system.

D. According to Padmanabhan and Chen, the brain can toggle between capturing olfactory information at a given moment and tracking changes in that information over time.

Question ID: 3dcc7140

Question

While researching a topic, a student has taken the following notes:

- Nissologists are scientists who study islands.
- Some nissologists define an island as any piece of land surrounded by water.
- Using that definition, they determined that Sweden has 221,000 islands.
- Other nissologists define an island as being 1 kilometer square, a certain distance from the mainland, and having at least 50 permanent residents.
- Using that definition, they determined that Sweden has 24 islands.

The student wants to make and support a generalization about nissologists' definition of an island. Which choice most effectively uses relevant information from the notes to accomplish these goals?

Answer

A. The definition of an island as any piece of land surrounded by water is supported by some nissologists, scientists who study islands.

B. Multiple counts of Sweden's islands have been based on different definitions of an island.

C. Based on a recent count, Sweden has a relatively small number of islands with at least 50 permanent residents.

D. Nissologists' different definitions can result in huge disparities in counts of islands, as the example of Sweden shows.

Question ID: 622a351d

Question

While researching a topic, a student has taken the following notes:

- In 1978, Sámi activists staged protests to block the construction of a dam on the Alta River in Norway.
- The dam would disrupt Sámi fishing and reindeer herding.
- The dam was ultimately built, but the Alta conflict had a lasting impact.
- It brought international attention to the issue of Sámi rights.
- It led to a set of 2005 legal protections establishing Sámi rights to lands, waters, and resources.

The student wants to make and support a generalization about the Alta conflict. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

A. During the Alta conflict, Sámi activists staged protests to block the construction of a dam on the Alta River in Norway that would disrupt local fishing and reindeer herding.

B. Although the dam that the Sámi activists had protested was ultimately built, the Alta conflict had a lasting impact.

C. Sámi rights to lands, waters, and resources received international attention and legal protections as a result of the Alta conflict.

D. The Alta conflict had a lasting impact, resulting in international attention and legal protections for Sámi rights to lands, waters, and resources.

Question ID: 5645f119

Question

While researching a topic, a student has taken the following notes:

- In a 2023 study, environmental scientist Jazmin Locke-Rodriguez and colleagues tested the use of floating treatment wetlands (FTWs) in Florida.
- FTWs are artificial floating platforms of plants used to remediate polluted or nutrient-imbalanced water.
- Finding: FTWs using marigold flowers removed 52% more total phosphorus than the control.
- Finding: The test yielded 65 market-quality blooms per square meter.
- The authors concluded marigolds showed “promising potential as a commercially viable remediating crop cultivated on FTWs in South Florida.”

The student wants to present the findings of the study. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

A. The authors concluded that marigolds grown on FTWs were “commercially viable,” having produced 65 blooms per square meter of market-quality blooms in a 2023 study.

B. In a 2023 study, Locke-Rodriguez and colleagues found that marigolds cultivated on FTWs produced 52% more market-quality flower blooms than the control.

C. Locke-Rodriguez and colleagues found that FTWs using marigolds not only helped remove phosphorus from the water but also yielded market-quality blooms.

D. FTWs using marigolds, Locke-Rodriguez and colleagues found, yielded 65 flower blooms and removed 52% of phosphorus from the water.

Question ID: 87d34a39

Question

While researching a topic, a student has taken the following notes:

- The National Congress of American Indians (NCAI) was founded in 1944 by representatives of fifty tribal governments.
- The NCAI was created to protect the sovereignty of Indigenous tribes.
- Napoleon B. Johnson (Cherokee) was the NCAI's first president.
- In 1975, the US Congress passed the Indian Self-Determination and Education Assistance Act (Public Law 96-638).
- This legislation formally acknowledged tribes' right to self-governance.
- The advocacy of the NCAI was a key factor in the law's passing.

The student wants to identify an accomplishment of the NCAI. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

A. The NCAI, founded by representatives of fifty tribal governments, had Napoleon B. Johnson (Cherokee) as its first president.

B. Founded in 1944, the NCAI was created by representatives of tribal governments from fifty sovereign Indigenous tribes.

C. The NCAI's advocacy was key to the passing of Public Law 96-638, legislation formally acknowledging Indigenous tribes' right to self-governance.

D. In 1975, the NCAI passed the Indian Self-Determination and Education Assistance Act, which was created to protect the sovereignty of Indigenous tribes.

Question ID: b0620764

Question

While researching a topic, a student has taken the following notes:

- Phobetor, a name drawn from Greek mythology, is an exoplanet that orbits the star PSR B1257+12, also known as Lich.
- Phobetor's mass is 0.01 times that of Jupiter, or 0.01 Jupiter masses.
- Mastika, which means "gem" or "jewel" in Malay, is an exoplanet that orbits the star HD 179949, also known as Gumala.
- Mastika's mass is 0.92 Jupiter masses.

The student wants to make and support a generalization about exoplanets. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

A. Exoplanets that are named Phobetor orbit Lich, and those that are named Mastika orbit Gumala.

B. Even though Phobetor and Mastika are both exoplanets, their masses are different: Phobetor's mass is 0.01 Jupiter masses, and Mastika's is 0.92 Jupiter masses.

C. Many stars have both a designation and a proper name; for instance, PSR B1257+12 is also known as Lich, and HD 179949 is also known as Gumala.

D. Exoplanet names have diverse origins, a fact that can be seen in the cases of Phobetor, a name drawn from Greek mythology, and Mastika, which means "gem" or "jewel" in Malay.

Question ID: c34d6bff

Question

While researching a topic, a student has taken the following notes:

- African American women played prominent roles in the Civil Rights Movement, including at the famous 1963 March on Washington.
- Civil rights activist Anna Hedgeman, one of the march's organizers, was a political adviser who had worked for President Truman.
- Civil rights activist Daisy Bates was a well-known journalist and advocate for school desegregation.
- Hedgeman worked behind the scenes to make sure a woman was included in the lineup of speakers at the march.
- Bates was the sole woman to speak, delivering a brief but memorable address to the cheering crowd.

The student wants to compare the two women's contributions to the March on Washington. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

A. Hedgeman and Bates contributed to the march in different ways; Bates, for example, delivered a brief but memorable address.

B. Hedgeman worked in politics and helped organize the march, while Bates was a journalist and school desegregation advocate.

C. Although Hedgeman worked behind the scenes to make sure a woman speaker was included, Bates was the sole woman to speak at the march.

D. Many African American women, including Bates and Hedgeman, fought for civil rights, but only one spoke at the march.

Question ID: db3ad406

Question

While researching a topic, a student has taken the following notes:

- Stars form in a galaxy when gravity causes a massive cloud of dust and gas to collapse.
- A galaxy in a phase of rapid star formation is called a starburst galaxy.
- Quenching is a process in which a galaxy loses star-forming gas.
- A galaxy that no longer forms stars is called a quenched galaxy.
- A quenched galaxy has entered the poststarburst phase.

The student wants to explain what a quenched galaxy is. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

A. Before quenching, a starburst galaxy will form stars at a rapid rate.

B. When it becomes quenched, a starburst galaxy enters the poststarburst phase.

C. Having entered the poststarburst phase, a quenched galaxy is one that no longer forms stars.

D. A starburst galaxy will lose star-forming gas and eventually become quenched.